

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 | SUB-STRAND 1.5: MOMENTS OF EQUILIBRIUM | FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Welcome to your Final Explanation! Throughout this unit you have investigated the Toppling Jiko Stool: a three-legged stool stood firm under a sufuria of ugali, but toppled when a full 20-litre jerrycan (weight 200 N) was placed at one corner and nudged, while a wider, lower four-legged stool with the same jerrycan at its corner only rocked and returned upright. Use your evidence from Lessons 1 to 6, your beam-balance and metre-rule data, your centre of gravity investigations, your cumulative model and your Driving Question Board to answer each prompt below. Write in full sentences, use correct physics words, show every step of any calculation with units, and refer to evidence you actually observed. Draw neat labelled diagrams where asked. Take your time, and think about how each answer helps answer our Driving Question: Why do some objects stay balanced while others topple, and how can we use the physics of turning forces to predict and control whether an object will be stable or fall?

Part 1: Turning Effects and Moments

In Lessons 1 and 2 you saw that the jerrycan caused toppling at the corner of the Jiko stool but did not cause it when the load was placed elsewhere. (a) What is a moment of a force, and what is the formula and SI unit used to calculate it? (b) Using your metre-rule and pencil-pivot activity, describe what you observed when you moved the 500 g mass further from the pivot, and explain why this happened even though the mass did not change. (c) Suppose the 200 N jerrycan acts at a perpendicular distance of 0.30 m from a pivot leg. Calculate its moment about that leg. Then calculate the moment if the jerrycan were moved closer to the pivot, to a perpendicular distance of 0.10

(a) A moment is the turning effect of a force about a pivot. It is calculated using $M = F \times d$, where F is the force in newtons (N) and d is the perpendicular distance from the line of action of the force to the pivot in metres (m). The SI unit is the newton-metre (N·m). (b) When I moved the 500 g mass further from the pencil pivot, the rule tipped more easily and was harder to balance. The mass, and therefore its weight (the force), stayed the same, but the perpendicular distance d increased, so the moment $M = F \times d$ increased. The turning effect depends on both the force and the distance, not on the force alone. (c) At $d = 0.30$ m: $M = F \times d = 200 \text{ N} \times 0.30 \text{ m} = 60 \text{ N}\cdot\text{m}$. At $d = 0.10$ m: $M = 200 \text{ N} \times 0.10 \text{ m} = 20 \text{ N}\cdot\text{m}$. The same 200 N force gives a moment three times larger when the distance is three times larger. This shows that the position of the jerrycan matters: the weight does not change, but the perpendicular distance from the pivot does, so the moment changes. A jerrycan placed far from the pivot leg, such as at the corner of the Jiko stool, gives a large turning effect compared with one placed nearer the pivot.

m. What does this comparison tell you about why the position of the jerrycan matters?	
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Part 2: Centre of Gravity and the Tipping Point

In Lesson 3 you used a plumb line and cardboard cut-outs, and tested blocks to see when they topple. (a) Define centre of gravity and describe how you found it for an irregular cardboard shape. (b) Using what you saw with the tall narrow block and the wide low block, state the condition that decides whether an object topples. (c) Explain, using centre of gravity, why the three-legged Jiko stool toppled when the jerrycan was placed at its corner. (d) Name the three states of equilibrium. Then decide which state the wider four-legged stool was in when it was nudged with the jerrycan at its corner, and give a reason based on what you observed.	(a) The centre of gravity is the single point through which the whole weight of an object appears to act. I found it by hanging the cardboard shape from different points with a plumb line and marking the vertical line each time; the lines crossed at one point, which is the centre of gravity. (b) An object topples when the vertical line through its centre of gravity falls outside its base of support. If the line stays inside the base, the object does not topple. (c) The jerrycan at the corner shifted the combined centre of gravity of the stool-plus-load outward and kept it high. With a small nudge, the vertical line through this combined centre of gravity moved outside the triangular base of support formed by the three legs. Then there was an unbalanced moment and no restoring moment, so the stool toppled. (d) The three states are stable, unstable and neutral equilibrium. The wider four-legged stool was in stable equilibrium: when it was nudged it only rocked and then returned to its upright position, because the vertical line through the centre of gravity stayed inside its base of support and a restoring moment brought it back.
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Part 3: The Principle of Moments

In Lesson 4 you balanced a metre rule on a pencil pivot (like a bao plank on a log) using bags of beans and coins and recorded clockwise and anti-clockwise moments. (a) State the Principle of Moments. (b) A bao plank is balanced on its pivot. On the left, a child's weight of 300 N sits 1.0 m from the pivot. On the right, a second child sits 1.5 m from the pivot. Calculate the weight of the	(a) When an object is balanced (in equilibrium), the sum of the clockwise moments about the pivot equals the sum of the anti-clockwise moments about the same pivot. (b) Anti-clockwise moment = $300 \text{ N} \times 1.0 \text{ m} = 300 \text{ N}\cdot\text{m}$. At balance, clockwise moment = anti-clockwise moment, so $W \times 1.5 \text{ m} = 300 \text{ N}\cdot\text{m}$. $W = 300 \text{ N}\cdot\text{m} \div 1.5 \text{ m} = 200 \text{ N}$. The second child must weigh 200 N. (c) When a stool tilts, it turns about the edge of its base on the side of the jerrycan, the tipping edge. The jerrycan's weight on that side gives a toppling moment about the edge. The weight of the stool acting through its centre of gravity gives a moment in the opposite direction, a restoring moment. If the restoring moment is at least as large as the toppling moment, the moments balance or return the stool upright; if the toppling moment is larger, there is a net moment and the stool topples. The wider four-legged stool has a larger base, so its tipping edge is further from its centre of gravity, which means the weight of the stool has a longer perpendicular distance from the edge and gives a larger restoring moment. With the jerrycan at the corner, its restoring moment was enough to bring it back, so it rocked and returned upright. The narrow three-legged stool has its tipping edge closer to its centre of gravity, so its restoring moment is smaller. With the jerrycan at the corner, the toppling moment was larger than the restoring moment, so the stool toppled. This agrees with the centre of gravity idea: the load shifts the combined centre of gravity, and where its vertical line falls relative to the tipping edge decides which moment is larger.
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second child needed for balance, showing your working. (c) A stool begins to tip about the edge of its base nearest the jerrycan. Explain, using moments about that tipping edge, what decides whether the stool topples or returns upright, and use this to explain why the wide four-legged stool did not topple with the same jerrycan at its corner but the narrow three-legged stool did.	
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Part 4: Stability in Design

In Lesson 5 you tested tall cardboard boxes with clay and widened the base with flaps, and analysed a crane incident in Nairobi. (a) Describe what you observed about the critical tipping angle when clay was at the top of the box compared with the base, and when the base was widened. (b) State two ways to make an object more stable and explain each using the idea of centre of gravity and base of support. (c) Explain why the crane in the case study toppled. (d) Name one real structure or career where engineers or workers use these ideas and describe how.	(a) With clay at the top of the box, the critical tipping angle was small, so the box toppled easily. With the clay at the base, the critical tipping angle was much larger. When cardboard flaps widened the base, the critical tipping angle increased even further. (b) First, lower the centre of gravity: a low centre of gravity means the box must be tilted further before the vertical line through it leaves the base. Second, widen the base of support: a wider base means the vertical line must travel further before it falls outside. Both make the object harder to topple. (c) The crane toppled because its load was swung beyond the rated radius. This shifted the effective centre of gravity of the crane-plus-load outside the base of support, producing an unbalanced toppling moment. (d) Double-decker buses are designed with heavy parts low down and a wide wheelbase so the centre of gravity stays low and inside the base even on a tilt. Crane operators keep loads within the rated radius so the moment of the load does not exceed the stabilising moment.
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Part 5: Explaining and Predicting with the Jiko Stool

Now bring everything together to answer the Driving Question. (a) Write a complete explanation of why the three-legged Jiko stool toppled with the 200 N jerrycan at its corner	(a) The jerrycan's 200 N weight at the corner acts at a large perpendicular distance from the tipping edge, so it creates a large turning moment ($M = F \times d$). It also shifts the combined centre of gravity of the stool-plus-load outward and keeps it high. While the vertical line through the combined centre of gravity is inside the base of support, the total weight gives a restoring moment about the tipping edge that pushes the stool back upright. When the nudge moves that vertical line outside the triangular base of support formed by the three legs, the total weight now acts on the far side of the tipping edge, so its moment turns the stool in the toppling direction. There is no restoring moment left, so the moments are unbalanced and the stool rotates and topples. The wider four-
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while the wider four-legged stool only rocked and returned upright. Link the ideas you used across the unit into one cause-and-effect chain, and explain both stools. (b) Predict what would happen if the jerrycan were placed at the centre of the three-legged stool, and justify with physics. (c) Suggest two changes a Kenyan market trader could make to a stool or a stack of goods, such as sacks of maize or crates, to stop it toppling, and explain why each works. (d) How does this unit show that Physics is relevant in daily life and that forces determine the stability of everyday objects?	legged stool has a larger base of support, so the vertical line through the combined centre of gravity stays inside the base even with the jerrycan at the corner. The weight still gives a restoring moment about the tipping edge, so the stool rocks and returns upright, which is stable equilibrium. (b) It would stay upright. The jerrycan's distance from the tipping edge would be small, so its toppling moment is small, and the combined centre of gravity stays inside the base of support, so the stool is in equilibrium. (c) First, put the heaviest items, such as full sacks of maize, at the bottom to lower the centre of gravity, so the box or stack must tilt further before the vertical line leaves the base. Second, use a wider base, for example spread the legs or place the stack on a broad crate, so the vertical line has further to travel before leaving the base. (d) Physics helps us predict and control whether everyday objects stay balanced, from stools and market stalls to buses and cranes. Forces create turning effects (moments), and the position of the centre of gravity relative to the base of support determines stability. By using moments and the principle of moments we can calculate and design for safety.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Moment calculations and the Principle of Moments	Correctly states $M = F \times d$ and the Principle of Moments, calculates all values with correct working and units (N·m, N), and interprets what the results mean for the stool.	States the formulas and gets most calculations right, with minor errors in units or working, and gives limited interpretation.	Gives incomplete or incorrect formulas, calculation errors, or missing units, with little or no interpretation.
Use of centre of gravity and base of support	Accurately defines centre of gravity and states the toppling condition (vertical line outside the base of support), and applies it precisely to both stools and the crane.	Defines the terms and states the condition mostly correctly, applying it to the stool with some gaps.	Confuses centre of gravity with other ideas, or does not link the vertical line to the base of support.
Evidence from the investigations	Refers specifically to observations from the lessons (500 g mass on the metre rule, plumb line, clay at top or base, flaps, 200 N jerrycan, crane case) to support every claim.	Refers to some observations from the lessons but some claims are not backed by evidence.	Makes general statements with little or no reference to what was observed.
Causal explanation of the anchoring phenomenon	Gives a complete cause-and-effect chain linking moments, shift of the combined centre of gravity, the base of support and equilibrium to explain both stools, and	Explains the phenomenon using most key ideas but the chain of reasoning has a missing link.	Describes what happened without explaining why, or uses ideas incorrectly.

	makes a correct prediction.		
Application and communication	Gives realistic Kenyan applications with correct reasoning, uses scientific vocabulary correctly throughout, and presents clear, well-organised answers.	Gives reasonable applications and mostly correct vocabulary, with some unclear parts.	Applications are missing or unrealistic, vocabulary is often incorrect, and answers are hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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